

QUANG MINH NGUYEN

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Education

Florida State University

Bachelor of Science in Computer Science (GPA: 3.9)

Expected May 2027

Tallahassee, Florida

Relevant Coursework: Object-oriented Programming(OOP), Data Structures and Algorithms, Computer Architecture, Database Management Systems, Software Engineering

Experience

Viet Tech Mentorship Program

Software Engineer Fellow

April 2025 - Present

Remote

- Developed Treverse, a full-stack application using **MongoDB, Express, NodeJS, ReactJS** in team of 10, enabling users to manage and track job applications, share interviews, and search postings efficiently.
- Collaborated closely with Product Designer to build an intuitive and appealing user interface using **Tailwind CSS** and **ShadCN**, ensuring a seamless and accessible user experience across devices.
- Implemented cursor-based pagination using **MongoDB aggregation pipeline**, improving retrieval time 50%, and integrated **TanStack Table** for dynamic filter state management and efficient rendering of large job posting datasets.
- Engineered an interview sharing feature that allows user to share interview experience and integrated **Vercel AI SDK** to call **Gemini API** for retrieving company details and generating insights based on shared interview content.
- Set up a daily **cron job** on **AWS Lambda** and deployed as a **Docker container** to automatically refresh interview insights, and utilized **Redis** to cache results for improved performance.

Florida Resources and Environmental Analysis Center

Application Developer Intern

Feb 2025 - Present

Tallahassee, Florida

- Engineered a GIS tool in **C++** that integrates Leidos' open-source **GSF (Generic Sensor Format)** library to decode sonar data and extract back scatter information for marine analysis.
- Reduced point density analysis runtime by 90% (from 13 days to 2 days) by identifying bottlenecks using **Cprofile** and optimizing geospatial workflows for seafloor mapping using in-memory processing and optimizing queries in **ArcPy**.
- Developed a document processing pipeline that parsed 20,000 corner zone PDFs, extracting geospatial coordinates using **PyTesseract** and **Gemini API**.

FPT Software Company

Software Engineer Intern

July 2024 – Aug 2024

Hanoi, Vietnam

- Developed a high-performance comment API using **Java** and **Spring Boot**, leveraging microservices architecture, asynchronous programming, and dependency injection to reduce response times by 30% and enhance client satisfaction.
- Collaborated in an Agile development team using Scrum methodology, participating in daily stand-ups, sprint planning, and bi-weekly retrospectives to ensure iterative delivery of features.

Projects

Grocery Trak | *Flutter, Python, Android Studio, Mobile Development*

Feb 2025 - March 2025

- Developed an AI-powered food recognition system using **OpenCV** and deep learning, enabling real-time scanning of grocery items to extract nutritional information and ingredients.
- Designed and implemented a comprehensive program to collect, merge, and validate up to 100,000 images from 10 datasets featuring 503 unique ingredients, enhancing the **YOLOv11-based** food recognition system's accuracy to 91%.

Film Rental System | *Spring, Hibernate, MySQL, Docker, Bootstrap, JQuery*

July 2024 - Aug 2024

- Engineered a comprehensive film rental system using the **Spring Framework**, delivering robust backend functionality that streamlined the movie renting process.
- Integrated user authentication with **Spring Security**, ensuring stringent data protection and authorized access.

Technical Skills

Languages: Python, Java, C++, HTML/CSS, JavaScript, TypeScript, SQL, Assembly

Technologies/Frameworks: Unix, Conda, GitHub, Spring, React, Angular, Hibernate, Docker

Skills: Data Structure, Problem-Solving Algorithms, System Design, Web Development(Full-Stack), Frontend, Backend